



DEPARTMENT OF AGRICULTURE

Inspection and Consumer Services Division

STORAGE AND HANDLING OF ANHYDROUS AMMONIA

8 CCR 1202-5

[Editor's Notes follow the text of the rules at the end of this CCR Document.]

SECTION 1. INTRODUCTION

1.1. Scope

- 1.1.1. These Rules apply to the design, construction, location, installation, and operation of anhydrous ammonia systems for storing, handling, transporting, and utilizing anhydrous ammonia as an agricultural fertilizer. They do not supersede any other Federal Statutes or Regulations not cited herein that may be applicable.
- 1.1.2. These Rules do not apply to:
 - 1.1.2.1. Ammonia, manufacturing plants.
 - 1.1.2.2. Refrigeration plants where ammonia is used solely as a refrigerant.
 - 1.1.2.3. Ammonia transportation pipelines.
 - 1.1.2.4. Refrigerated ammonia systems.

1.2. General

- 1.2.1. Where the provisions of these Rules impose undue hardship or where the literal adherence to the provisions fails to provide adequate safety, the Commissioner may permit deviation from the Rules.
- 1.2.2. The term "anhydrous ammonia" as used in these Rules refers to the compound, formed by a combination of two gaseous elements, nitrogen and hydrogen, in the proportion of one part nitrogen to three parts hydrogen by volume. Anhydrous ammonia may be in either gaseous or liquid form. It is not to be confused with aqua ammonia which is a solution of ammonia gas in water. Whenever the term "ammonia" appears in these Rules, it means anhydrous ammonia.
- 1.2.3. It is important that personnel understand the properties of this gas and that they be thoroughly trained in safe practices for its storage and handling. Some of the important physical properties of ammonia are listed in 1.2.5.
- 1.2.4. Gaseous ammonia liquefies under pressure at ambient temperature. Advantage of this characteristic is taken by industry and for convenience this commodity is usually shipped and stored under pressure as a liquid. When refrigerated to or below its normal boiling point (-28°F) it may be shipped and stored as a liquid at atmospheric pressure.

1.2.5. Physical Properties of Ammonia:

Molecular symbol	NH ₃
Molecular weight	17.032
Boiling point at one atmosphere*	-28°F
Melting point at one atmosphere*	-107.9°F
Critical temperature	271.4°F
Critical pressure	1657 psia.
Latent heat at -28°F and one atmosphere*	589.3 Btu per pound
Relative density of vapor compared to dry air at 32°F and one atmosphere*	0.5970
Vapor density at -28°F and one atmosphere*	0.05555 lb. per cu ft
Specific gravity of liquid at -28°F compared to water at 39.2°F	0.6819
Liquid density at -28°F and one atmosphere*	42.57 lb. per cu ft
Specific volume of vapor at 32°F and one atmosphere*	20.78 cu ft per pound
Flammable limits by volume in air at atmospheric pressure	16% to 25%
Ignition Temperature (in a standard quartz container)	1562°F
Specific Heat, Gas, 15°C, one atm at constant pressure, C _p	0.5232 Btu/lb. °F
at constant volume, C _v	0.3995 Btu/lb. °F

*One atmosphere = 14.7 psia.

- 1.2.6. Experience has shown that ammonia is extremely hard to ignite and under normal conditions is a very stable compound. It takes temperatures of 840-980°F to cause it to dissociate slightly at atmospheric pressure. The flammable limits at atmospheric pressure are 16% to 25% by volume of ammonia in air. Experiments conducted by a nationally recognized laboratory indicated that an ammonia-air mixture in a standard quartz test container does not ignite below 1562°F. Ammonia is classified by the United States Department of Transportation and the United States Coast Guard as a non-flammable compressed gas for the purpose of transportation.
- 1.2.7. Ammonia should be handled only by properly trained personnel. In no case should ammonia be used in conjunction with chemicals unless the possible reactions have first been adequately investigated. Under some circumstances ammonia and ammonium compounds can form explosive products with other chemicals.
- 1.2.8. Ammonia gas irritates the skin and mucous membrane. Its odor is detectable by most people. Since ammonia gas is lighter than air, adequate ventilation is the best means of preventing any accumulation.
- 1.2.9. Zinc, copper and copper base alloys such as brass are subject to rapid destructive action by ammonia in the presence of water.

1.3. Standards Incorporated by Reference

- 1.3.1. The following standards are hereby incorporated by reference in these Rules:

- 1.3.1.1. U.S. Department of Transportation (DOT) Regulations referring to Hazardous Materials Regulations of the Department of Transportation, including specifications for shipping containers, Code of Federal Regulations, 49 CFR parts 171 to 190, October 1, 2011 edition. This rule does not adopt any later amendments or editions.
- 1.3.1.2. American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code parts UW-1 through UW-65, Section VIII, Division 1, Titled "Part UW: Requirements for Pressure Vessels Fabricated by Welding," 2015 edition. This rule does not adopt any later amendments or editions.

- 1.3.1.3. Unfired Pressure Vessel Code of the American Society of Mechanical Engineers (Sections VIII of the ASME Boiler Construction Code), 1952, 1956, 1959, 1962, 1965, 1968, 1971, 1978, 1983, 1989, 1996, 1997, 2001, 2007, 2008, 2010 and 2015 editions. This rule does not adopt any later amendments or editions.
- 1.3.1.4. National Electrical Code, NFPA 70 (ANSI-C1), for Class I, Group D locations, 2014 edition. This rule does not adopt any later amendments or editions.
- 1.3.1.5. American Petroleum Institute, API Standard 620, Twelfth edition, 2013. This rule does not adopt any later amendments or editions.
- 1.3.1.6. Portable U.S. Department of Transportation (DOT) Containers, Code of Federal Regulations 49 CFR 173.32 October 1, 2011 edition. This rule does not adopt any later amendments or editions.
- 1.3.1.7. Compressed Gas Association Publication P-1, 2015 edition. This rule does not adopt any later amendments or editions.
- 1.3.1.8. U.S. Department of Transportation (DOT) Regulation referring to systems utilizing cylinders, portable tanks (DOT-51), Code of Federal Regulations 49 CFR Part 173.32, October 1, 2011 edition. This rule does not adopt any later amendments or editions.
- 1.3.1.9. U.S. Department of Transportation (DOT) Regulations referring to systems utilizing "Ton Containers" (DOT-106A, DOT-110A) Code of Federal Regulations 49 CFR Parts 179.300-179.301, October 1, 2011 edition. This rule does not adopt any later amendments or editions.
- 1.3.1.10. The American National Standards Institute's Method of Marking Portable Compressed Gas Containers to Identify the Material Contained, ANSI Z48.1, 1954 edition. This rule does not adopt any later amendments or editions.
- 1.3.1.11. U.S. Department of Transportation (DOT) Regulations referring to safety relief device requirements, Code of Federal Regulations 49 CFR Part 173.315, October 1, 2011 edition. This rule does not adopt any later amendments or editions.
- 1.3.2. A copy of the Official Publications of the above incorporated Standards shall be kept on file at the Colorado Department of Agriculture, located at 305 Interlocken Parkway, Broomfield Colorado 80021, and shall be open to public inspection, and available for copying, during normal business hours.

A copy of the official publication of the U.S. Department of Transportation (DOT) regulations shall also be kept at the U.S. Department of Transportation, 1200 New Jersey Avenue SE, Washington D.C. 20590 and is available online at: transportation.gov/regulations.

A copy of the official publication of the American Society of Mechanical Engineers shall be kept on file at the Colorado Department of Agriculture, located at 305 Interlocken Parkway, Broomfield Colorado 80021, and shall be open to public inspection, and available for copying, during normal business hours. A copy of the official publication of the American Society of Mechanical Engineers shall also be kept at the ASME Headquarters, Two Park Avenue, New York NY 10016-5990 and is available online at: ASME.org

A copy of the official publication of The National Electrical Code shall be kept on file at the Colorado Department of Agriculture, located at 305 Interlocken Parkway, Broomfield Colorado 80021, and shall be open to public inspection, and available for copying, during normal business hours. A copy of the official publication of the National Electrical Code shall also be kept at The National Fire Protection Association, 1 Batterymarch Park, Quincy Massachusetts 02169-7471 and is available online at: NFPA.org

A copy of the American Petroleum Institute Standards shall be kept on file at the Colorado Department of Agriculture, located at 305 Interlocken Parkway, Broomfield Colorado 80021, and shall be open to public inspection, and available for copying, during normal business hours. A copy of the official publication of the American Petroleum Institute Standards shall also be kept at the American Petroleum Institute, 200 Massachusetts Avenue NW, suite 1100, Washington D.C., 20001-5571 and is available online at: API.org.

A copy of the Compressed Gas Association Publication shall be kept on file at the Colorado Department of Agriculture, located at 305 Interlocken Parkway, Broomfield Colorado 80021, and shall be open to public inspection, and available for copying, during normal business hours. A copy of the official publication of the Compressed Gas Association shall also be kept at 14501 George Carter Way, suite 103, Chantilly, VA 20151 and is available online at: CGANET.com.

A copy of the American National Standards Institute's standards Publication shall be kept on file at the Colorado Department of Agriculture, located at 305 Interlocken Parkway, Broomfield Colorado 80021, and shall be open to public inspection, and available for copying, during normal business hours. A copy of the official publication of the American National Standards Institute's Standards shall also be kept at ANSI headquarters, 1899 L Street NW, Washington D.C. 20036 and is available online at: ANSI.org.

1.4. Definitions

The following definitions are used for the purpose of these Rules:

1.4.1. "Approved" means:

1.4.1.1. Listed by a recognized testing laboratory, or

1.4.1.2. Recommended by the manufacturer as suitable for use with anhydrous ammonia and so marked, or

1.4.1.3. Accepted by the Commissioner.

1.4.2. "Appurtenance" refers to all devices such as safety relief devices, liquid level gauging devices, valves, pressure gauges, fittings, metering or dispensing devices.

1.4.3. "ASME" means American Society of Mechanical Engineers.

1.4.4. "Backflow check valve" means a device designed to prevent ammonia from flowing in the wrong direction within a pipe or tube.

1.4.5. "Capacity" refers to the total volume of the container measured in U.S. gallons, unless otherwise specified.

1.4.6. "Chemical-splash goggles" and "splash proof goggles" means flexible-fitting chemical-protective goggles, with a hooded, indirect ventilation system that provides protection to the eyes and eye sockets from the splash of hazardous liquids. This term shall not include direct vented goggles.

- 1.4.7. "Cylinder" means a container of 1000 pounds water capacity or less.
- 1.4.8. The "Code" refers to the Unfired Pressure Vessel Code of the American Society of Mechanical Engineers (Section VIII of the ASME Boiler Construction Code), 2015 edition.
- 1.4.9. "Container" includes all vessels, tanks, cylinders or spheres used for transportation, storage or application of anhydrous ammonia.
- 1.4.10. "Data plate" means a piece of non-corroding metal permanently attached by the manufacturer to the surface of a container.
- 1.4.11. "Densely populated area" means any location with either one or more multifamily housing units or eight or more single-family dwellings located within a quarter section.
- 1.4.12. "Design Pressure" is identical to the term "Maximum Allowable Working Pressure" used in the Code.
- 1.4.13. "Emergency shutoff valve" means a valve that stops the flow of product by spring closure, gravity, or pressure and can be activated by an outside means including a cable pull, hose pull, air assists, electrical closure or back pressure.
- 1.4.14. "Excess-flow valve" means a device placed in a line that is designed to close when the flow of vapor or liquid flowing through the line exceeds the amount for which the valve is rated.
- 1.4.15. An "Implement of Husbandry" is a farm wagon-type tank vehicle of not over 3000 gallons capacity, used as a field storage "nurse tank" supplying the fertilizer to a field applicator and moved on highways for bringing the fertilizer from a local source of supply to farms or fields or from one farm or field to another.
- 1.4.16. "Filling Density" means the percent ratio of the weight of the gas in a container to the weight of water at 60°F that the container will hold. For determining the water capacity of the tank in pounds, the weight of a gallon (231 cubic inches) of water at 60°F in air shall be 8.32828 pounds.
- 1.4.17. "Gas" refers to anhydrous ammonia in either the gaseous or liquefied state.
- 1.4.18. "DOT Regulations" refer to Hazardous Materials Regulations of the U.S. Department of Transportation (Title 49-Transportation, Code of Federal Regulations, Parts 171 to 190), including Specifications for Shipping Containers.
- 1.4.19. "Mobile container" means any container that is not installed as a permanent storage container.
- 1.4.20. "Non-code welding" means welding that does not comply with parts UW-1 through UW-65 of the ASME Boiler and Pressure Vessel Code, Sections VIII, Division 1, Titles "Parts UW: Requirements for Pressure Vessels Fabricated by Welding," 2015 edition.
- 1.4.21. "Permanent storage container" means a stationary container having a volume of at least 3,000 water gallons.
- 1.4.22. "Permanent storage facility" means a site that includes one or more permanent storage containers and their connections and appurtenances.
- 1.4.23. "Place of public assembly" means any building, structure or facility established to accommodate groups of people for commercial, civic, political, religious, recreational, educational, or other purposes. This term shall include buildings or structures used for medical care, including hospitals, assisted care facilities, and prisons.

- 1.4.24. "Systems" as used in these Rules refers to an assembly of equipment consisting essentially of the container or containers, appurtenances, pumps, compressors, and interconnecting piping.
- 1.4.25. The abbreviations "psig" and "psia" refer to pounds per square inch gauge and pounds per square inch absolute, respectively.
- 1.4.26. The terms "charging" and "filling" are used interchangeably and have the same meaning.
- 1.4.27. "Trailer" as used in these Rules refers to every vehicle designed for carrying persons or property while being drawn by a motor vehicle and so constructed that no part of its weight except the towing device rests upon the towing vehicle.
- 1.4.28. "Tank Motor Vehicle" means any motor vehicle designed or used for the transportation of anhydrous ammonia in any tank designed to be permanently attached to any motor vehicle or any container not permanently attached to any motor vehicle which by reason of its size, construction or attachment to any motor vehicle must be loaded and/or unloaded without being removed from the motor vehicle.
- 1.4.29. "Semi-trailer" refers to every vehicle designed for carrying persons or property while being drawn by a motor vehicle and so constructed that some part of its weight and that of its load rests upon or is carried by another vehicle.
- 1.4.30. "Safety Relief Valve" refers to an automatic spring loaded or equivalent type pressure activated device for gas or vapor service characterized by pop action upon opening, sometimes referred to as a pop valve.
- 1.4.31. "Hydrostatic Relief Valve" refers to an automatic pressure activated valve for liquid service characterized by throttle or slow weep opening (non-pop action).

SECTION 2. DESIGN OF ANHYDROUS AMMONIA SYSTEMS

2.1. Painting of Containers

- 2.1.1. Above-ground uninsulated containers shall have a reflective surface maintained in good condition. White is recommended for painted surfaces, but other light reflecting colors are acceptable.

2.2. Electrical Equipment and Wiring

- 2.2.1. Where concentrations of ammonia in air in excess of 16% by volume are likely to be encountered, electrical equipment and wiring shall be of a type specified by and be installed in accordance with National Electrical Code, NFPA 70 (ANSI-C1), for Class I, Group D locations.

2.3. Marking on Containers and Systems other than DOT Containers

- 2.3.1. System nameplates, when required, shall be permanently attached to the system so as to be readily accessible for inspection and shall include markings as prescribed in 2.3.2.
- 2.3.2. Each container or system covered in Sections 7, 9, 10 and 11 shall be marked as follows:
- 2.3.2.1. With a marking identifying compliance with the Rules of the code under which the container is constructed.
- 2.3.2.2. With a notation on the container and system nameplate when the system is designed for underground installation.

- 2.3.2.3. With the date of fabrication and the name and address of the supplier of the container or the trade name of the container.
- 2.3.2.4. With the water capacity of the container in pounds at 600F or gallons, U.S. standard.
- 2.3.2.5. With the design pressure in pounds per square inch gauge.
- 2.3.2.6. With the wall thickness of the shell and heads.
- 2.3.2.7. With markings indicating the maximum level to which the container may be filled with liquid anhydrous ammonia at temperatures between 200F and 1000F except on containers provided with fixed maximum level indicators, such as fixed length dip tubes, or containers that are filled by weight.
- 2.3.2.8. With the outside surface area in square feet.
- 2.3.2.9. With minimum temperature in Fahrenheit for which the container is designed.
- 2.3.2.10. Marking as specified on containers shall be on the container itself or on a nameplate permanently affixed thereto.
- 2.3.3. All main operating valves on permanently installed containers having a capacity of over three thousand water gallons shall be identified to show whether the valve is in liquid or vapor service. The recommended method of identification may be legend or color code as specified in 2.3.3.1 or 2.3.3.2.
 - 2.3.3.1. Legend: The legend liquid (or liquid valve), vapor (or vapor valve), as appropriate, shall be placed on or within twelve inches of the valve by means of a stencil tag, or decal.
 - 2.3.3.2. Color code: Liquid valves shall be painted orange and vapor valves shall be painted yellow. The legend orange-liquid, yellow-vapor shall be displayed in one or more conspicuous places at each permanent storage location. The legend shall have letters at least two inches high and shall be placed against a contrasting background.

2.4. Container Appurtenances

- 2.4.1. All appurtenances shall be designed for not less than the maximum working pressure of that portion of the system on which they are installed.
 - 2.4.1.1. All appurtenances shall be fabricated from materials proved suitable for anhydrous ammonia service.
 - 2.4.1.2. Each valve and appurtenance shall be designed for not less than the maximum pressure to which the valve and appurtenance will be subjected.
 - 2.4.1.3. Each valve that could be subjected to container pressures shall have a rated working pressure of at least 250 psig.
 - 2.4.1.4. Connections to containers shall be limited to liquid-level gauges, emergency shutoff valves, pressure gauges, vapor-relief valves, liquid lines, vapor lines, and thermometers.
- 2.4.2. All connections to containers except safety relief devices, gauging devices, or those fitted with a No. 54 drill size orifice shall have shut-off valves located as close to the container as practicable.

- 2.4.3. The emergency shutoff valve shall be placed in the liquid line internally or externally to the container.
 - 2.4.3.1. If an external valve is used, the valve shall be located after the manual shutoff valve but as close to the opening of the container as possible.
 - 2.4.3.2. The emergency shutoff valve shall work properly from a remote location or when activated at the valve.
- 2.4.4. Excess flow valves where required by these Rules shall close automatically at the rated flows of vapor or liquid as specified by the manufacturer. The connections and line including valves and fittings being protected by an excess flow valve shall have a greater capacity than the rated flow of the excess flow valve.
- 2.4.5. Liquid level gauging devices that require bleeding of the product to the atmosphere and which are so constructed that outward flow will not exceed that passed by a No. 54 drill size opening need not be equipped with excess flow valves.
- 2.4.6. Openings from containers or through fittings attached directly on containers to which pressure gauge connections are made need not be equipped with excess flow valves if such openings are not larger than No. 54 drill size.
- 2.4.7. Excess flow and back pressure check valves where required by these Rules shall be located inside of the container or at a point outside as close as practicable to where the line enters the container. In the latter case, installation shall be made in such manner that any undue stress beyond the excess flow or back pressure check valve will not cause breakage between the container and the valve.
- 2.4.8. Excess flow valves shall be designed with a by-pass, not to exceed a No. 60 drill size opening to allow equalization of pressures.
- 2.4.9. Shut-off valves provided with an excess flow valve shall be designed for proper installation in a container connection so that the excess flow valve will close should the shut-off valve break.
- 2.4.10. All excess flow valves shall be plainly and permanently marked with the name or trademark of the manufacturer, the catalog number, and the rated capacity.
- 2.4.11. Each liquid connection used to fill a permanent storage container shall be fitted with a backflow check valve.
- 2.4.12. All appurtenances to any permanent storage container shall be protected from tampering and mechanical damage, including damage from vehicles. Each manually controlled valve that, if open, would allow ammonia to be transferred or released, shall be kept locked when unattended and during nonbusiness hours.

2.5. Piping, Tubing and Fittings

- 2.5.1. All piping, tubing and fittings shall be made of material suitable for anhydrous ammonia service.
- 2.5.2. All piping, tubing and fittings shall be designed for a pressure not less than the maximum pressure to which they may be subjected in service.
- 2.5.3. All piping shall be well supported and provision shall be made for expansion and contraction.

- 2.5.4. Piping used on non-refrigerated systems shall be at least ASTM A-53 Grade B Electric Resistance Welded and Electric Flash Welded Pipe or equal. Such pipe shall be at least Schedule 40 when joints are welded, or welded and flanged. Such pipe shall be at least Schedule 80 when joints are threaded. Brass, copper, or galvanized steel pipe or tubing shall not be used.
- 2.5.5. All metal flexible connections for permanent installations shall have a minimum working pressure of 250 psig (safety factor of 4). For temporary installations, hose meeting the requirement of 2.6 may be used.
- 2.5.6. Cast iron fittings shall not be used but this shall not prohibit the use of fittings made specifically for ammonia service of malleable or nodular iron such as specification ASTM A47 or ASTM A395.
- 2.5.7. Provisions shall be made for expansion, contraction, jarring, vibration, and for settling.
- 2.5.8. Adequate provisions shall be made to protect all exposed piping from physical damage that might result from moving machinery, the presence of automobiles or trucks, or any other undue strain that may be placed upon the piping.
- 2.5.9. Joint compounds shall be resistant to ammonia.
- 2.5.10. After assembly, all piping and tubing shall be tested and proved to be free from leaks at a pressure not less than the normal operating pressure of the system.

2.6. Hose Specification

- 2.6.1. Hoses used in ammonia service and subject to container pressure shall conform to the specification set forth in Appendix B of this Rule, which are based on The Rubber Manufacturers Association and The Fertilizer Institute "Hose Specifications for Anhydrous Ammonia".
- 2.6.2. Hose subject to container pressure shall be designed for a minimum working pressure of 350 psig and a minimum burst pressure of 1750 psig. Hose assemblies, when made up, shall be capable of withstanding a test pressure of 500 psig.
- 2.6.3. Hose and hose connections located on the low pressure side of flow control or pressure reducing valves on devices discharging to atmospheric pressure shall be designed for the maximum low side working pressure. All connections shall be designed, constructed, and installed so that there will be no leakage when connected.
- 2.6.4. Where liquid transfer hose is not drained of liquid upon completion of transfer operations, such hose shall be equipped with an approved shut-off valve at the discharge end. Provision shall be made to prevent excessive hydrostatic pressure in the hose. (See Rule 2.7.16.).
- 2.6.5. On all hose one-half inch O.D. and larger, used for the transfer of anhydrous ammonia liquid or vapor, there shall be etched, cast, or impressed at five-foot intervals the following information:
 - 2.6.5.1. "Anhydrous Ammonia";
 - 2.6.5.2. xxx psig (Maximum working pressure);
 - 2.6.5.3. Manufacturer's Name or Trademark;
 - 2.6.5.4. Year of Manufacture; and
 - 2.6.5.5. The date specified by the manufacturer on which the hose is to be removed from service (Manufacturer's removal date).

- 2.6.6. Each hose shall be replaced before or upon the expiration of the manufacturer's removal date.
- 2.6.7. A hose shall be removed from service prior to the manufacturer's removal date if a visual examination reveals any of the following:
- 2.6.7.1. Illegibility of any of the markings required in subsection 2.6.5;
 - 2.6.7.2. Cuts exposing reinforcing fabric;
 - 2.6.7.3. Soft spots or bulges in the hose;
 - 2.6.7.4. A blistering or loose outer covering;
 - 2.6.7.5. Kinking or flattening;
 - 2.6.7.6. Stretch marks;
 - 2.6.7.7. Slippage at any coupling; or
 - 2.6.7.8. Any other damage that could compromise the integrity of the safe use of the hose.

2.7. Safety Relief Devices

- 2.7.1. Every container used in systems covered by Sections 7, 9, 10 and 11 shall be provided with one or more safety relief valves of the spring-loaded or equivalent type. The discharge from safety relief valves shall be vented away from the container, upward and unobstructed to the atmosphere. All safety relief valve discharge openings shall have suitable rain caps that will allow free discharge of the vapor and prevent the entrance of water. Provision shall be made for draining condensate which may accumulate. The rate of the discharge shall be in accordance with the provisions of Appendix A of this Rule.
- 2.7.2. Container safety relief valves shall be set to start-to-discharge as follows, based on the design pressure of the container:

Containers	Minimum	Maximum*
ASME-U-68, U-69	110%	125%
ASME-U-200, U-201	95%	100%
ASME 1952, 1956, 1959, 1962, 1965, 1968 or 1971	95%	100%
API-ASME	95%	100%
U.S. Coast Guard	[As required by USCG regulations]	
DOT	[As required by DOT Regulations]	

* NOTE: A RELIEF VALVE MANUFACTURER'S TOLERANCE OF PLUS 10% IS PERMITTED.

- 2.7.3. Safety relief devices used in systems covered by Sections 7, 9, 10 and 11 shall be constructed to discharge at not less than the rates required in 2.7.1 before the pressure is in excess of 120% (not including the 10% tolerance referred to in 2.7.2) of the maximum permitted start-to-discharge pressure setting of the device.
- 2.7.4. Safety relief valves shall be so arranged to minimize the possibility of tampering. If the pressure setting adjustment is external, the relief valves shall be provided with means for sealing the adjustment.

- 2.7.5. Shut-off valves shall not be installed between the safety relief valves and the containers or systems described in Sections 7, 9, 10 and 11, except that a shut-off valve may be used where the arrangement of this valve is such as always to afford required capacity flow through the relief valves.

NOTE: The above exception is made to cover such cases as a three-way valve installed under two safety relief valves, each of which has the required rate of discharge and is so installed as to allow either of the safety relief valves to be closed off, but does not allow both safety valves to be closed off at the same time. Another exception to this may be where two separate relief valves are installed with individual shut-off valves. In this case, the two shut-off valve stems shall be mechanically interconnected in a manner which will allow full required flow of one safety relief valve at all times. Still another exception is a safety relief valve manifold which allows one valve of two, three, four or more to be closed off and the remaining valve or valves will provide not less than the rate of discharge shown on the manifold nameplate.

- 2.7.6. Safety relief valves shall have direct communication with the vapor space of the container.
- 2.7.7. Each safety relief valve used with systems described in Sections 7, 9, 10 and 11 shall be plainly and permanently marked as follows:

2.7.7.1. With the letters "AA" or the symbol "NH3".

2.7.7.2. The pressure in pounds per square inch gauge (psig) at which the valve is set to start-to-discharge.

2.7.7.3. The rate of discharge of the valve in cubic feet per minute of air at 60°F and atmospheric pressure (14.7 psia).

2.7.7.4. The manufacturer's name and catalog number.

For example, a safety relief valve marked AA-250-4200 (air) would mean that this valve is suitable for use on an anhydrous ammonia container; that it is set to start-to-discharge at 250 psig; and that its rate of discharge (see Sections 2.7.1, 2.7.2, and 2.7.3) is 4200 cubic feet per minute of air.

- 2.7.8. Each pressure-relief valve shall be manufactured for use with anhydrous ammonia and be installed, maintained, and replaced according to the manufacturer's instructions.
- 2.7.9. Unless otherwise specified by the manufacturer, a pressure-relief valve shall not be used for more than five years after the date of manufacture of the pressure relief device.
- 2.7.10. Each safety valve shall be replaced if the valve meets any of the following conditions:
- 2.7.10.1. Fails to meet applicable requirements;
 - 2.7.10.2. Shows evidence of damage, corrosion, or foreign matter; or
 - 2.7.10.3. Does not have functional weep holes that permit moisture to escape.
- 2.7.11. If moisture accumulation could occur in a vent, suitable provision shall be made to drain the moisture from the vent.
- 2.7.12. The flow capacity of the safety relief valve shall not be restricted by any connection to it on either the upstream or downstream side.

- 2.7.13. Vent pipes or tubing used to channel releases from safety relief valves shall not be restricted or smaller in size than the safety relief valve outlet connection.
- 2.7.14. Vent pipes may be connected and channeled into a common header if the cross-sectional area of the header is at least equal to the sum of the cross-sectional areas of each of the individual vent pipes.
- 2.7.15. The manufacturer or supplier of a safety relief valve manifold shall publish complete data showing the flow rating through the combined assembly of the manifold with safety relief valves installed. The manifold flow rating shall be determined by testing the manifold with all but one valve discharging. If one or more openings have restrictions not present in the remaining openings, the restricted opening or openings or those having the lowest flow shall be used to establish the flow rate marked on the manifold nameplate. The marking shall be similar to that required in 2.7.7 for individual valves.
- 2.7.16. A hydrostatic relief valve shall be installed between each pair of valves in the liquid ammonia piping or hose where liquid may be trapped so as to release into the atmosphere at a safe location.
- 2.7.17. Discharge from safety relief devices shall not terminate in or beneath any building.

2.8. Liquid Level Gauging Device

- 2.8.1. Each container except those filled by weight shall be equipped with an approved liquid level gauging device.
- 2.8.2. All gauging devices shall be arranged so that the maximum liquid level to which the container is filled is readily determined (See 2.3.2.7).
- 2.8.3. Gauging devices that require bleeding of the product to the atmosphere such as the rotary tube, fixed tube, and slip tube devices, shall be designed so that the maximum opening of the bleed valve is not larger than No. 54 drill size unless provided with an excess flow valve. (This requirement does not apply to farm vehicles used for the application of ammonia, as covered in Section 11.)
- 2.8.4. Gauging devices shall have a design pressure equal to or greater than the design pressure of the container on which they are installed.
- 2.8.5. Fixed liquid level gauges shall be so designed that the maximum volume of the container filled by liquid shall not exceed 85% of its water capacity. The coupling into which the fixed liquid level gauge is threaded must be placed at the 85% level of the container. If located elsewhere, the dip tube of this gauge must be installed in such a manner that it cannot be readily removed.
- 2.8.6. Gauge glasses of the columnar type shall be restricted to stationary storage installations. They shall be equipped with shut-off valves having metallic hand wheels, with excess-flow valves, and with extra heavy glass adequately protected with a metal housing applied by the gauge manufacturer. They shall be shielded against the direct rays of the sun.

SECTION 3. CONSTRUCTION OF ANHYDROUS AMMONIA SYSTEMS

3.1. Requirements for New Construction and Original Test of Containers, Including Skid Containers

- 3.1.1. Each container shall be constructed and tested in accordance with the Code and shall have a minimum design pressure of 250 psig.

- 3.1.2. Containers used with systems covered in Sections 7, 9, 10 and 11 shall be constructed and tested in accordance with the ASME Unfired Pressure Vessel Code except that construction under Table UW 12 at a basic joint efficiency of under 80% is not authorized.
 - 3.1.2.1. Containers built according to the Code do not have to comply with paragraphs UG 125 through UG 128 inclusive, and paragraphs UG 132 and UG 133.
- 3.1.3. Containers exceeding 36 inches in diameter or 250 gallons capacity shall be constructed to comply with one or more of the following additional requirements:
 - 3.1.3.1. Containers shall be stress relieved after fabrication in accordance with the Code, or
 - 3.1.3.2. Cold formed heads, when used shall be stress relieved, or
 - 3.1.3.3. Hot formed heads shall be used.
- 3.1.4. Welding to the shell, head, or any other part of the container subject to internal pressure shall be done in compliance with the Code under which the container was fabricated and shall be performed by a person or company that has a current certificate of authorization from the National Board of Boiler and Pressure Vessel Inspections. Other welding is permitted only on saddle plates, lugs, or brackets attached to the container by the container manufacturer.
- 3.1.5. All records of inspections and welding on the container shall:
 - 3.1.5.1. Be maintained by the owner of the container;
 - 3.1.5.2. Be made available to the Commissioner upon request; and
 - 3.1.5.3. Be transferred with change of ownership of the container.
- 3.1.6. All containers shall be inspected by a person having a current certificate of competency from the National Board of Boiler and Pressure Vessel Inspectors.
- 3.1.7. The provisions of 3.1.2 shall not be construed as prohibiting the continued use or reinstallation of containers constructed and maintained in accordance with the ASME Unfired Pressure Vessel Code.

SECTION 4. LOCATION OF ANHYDROUS AMMONIA SYSTEMS

4.1. Location of Containers

- 4.1.1. Consideration shall be given to the physiological effects of ammonia as well as to adjacent fire hazards in selecting the location for a storage container. Containers shall be located outside of buildings or in buildings or sections thereof approved for this purpose.
- 4.1.2. Containers shall be located at least 50 feet from a dug well or other sources of potable water supply, unless the container is a part of a water treatment installation.
- 4.1.3. The location of permanent storage containers shall be outside densely populated areas, unless the location has been approved in writing by the appropriate local governing body.
- 4.1.4. Container locations shall comply with the following table:

Normal Capacity of Container (Gallons)	Minimum Distances (feet) from Container to:		
	Line of Adjoining Property Which may be Built Upon, Highways & Mainline of Railroad	Place of Public Assembly	Institution Occupancy
Over 500 to 2,000	25	150	250
Over 2,000 to 30,000	50	300	500
Over 30,000 to 100,000	50	450	750
Over 100,000	50	600	1,000
Location approval required from Department within 30 days from notification.			

- 4.1.5. Storage areas shall be kept free of readily ignitable materials such as waste, weeds and long dry grass.

SECTION 5. INSTALLATION OF ANHYDROUS AMMONIA SYSTEMS

- 5.1. Each permanent storage container shall be mounted on either of the following:
- 5.1.1. A skid assembly with sufficient surface area to properly support the skid-mounted container or;
 - 5.1.2. Reinforced concrete footings and foundations or:
 - 5.1.3. Structural steel supports mounted on reinforced concrete foundations. The reinforced concrete foundations or footings shall extend below the established frost line and shall be constructed to support the total weight of the containers and their contents. If the container is equipped for bottom withdrawal, the container's foundation shall maintain the lowest point of the container at not less than 18 inches above ground level.
 - 5.1.4. Each container shall be mounted on its foundation in a manner that permits expansion and contraction. Each container shall be adequately supported so as to prevent the concentration of excessive loads on the supporting portion of the shell. Corrosion prevention measures shall be utilized on any portion of the container that is in contact with either the foundation or saddles.
 - 5.1.5. Each container shall be securely anchored.
- 5.2. Before installing or relocating a permanent storage container or permanent storage facility, the owner shall submit a detailed diagram of the permanent storage facility to the Commissioner for review to ensure that the proposed site meets the requirements in Section 4.
- 5.2.1. The name of the permanent storage facility and the telephone number to be contacted in case of an emergency shall be posted and be legible from each facility entrance using letters at least two inches high.
 - 5.2.2. No permanent storage container shall be located inside an enclosed structure unless the structure is specifically constructed for this purpose.

SECTION 6. OPERATION OF ANHYDROUS AMMONIA SYSTEMS

6.1. Use of Water in Emergencies

- 6.1.1. The concentration of ammonia vapor in air can effectively be reduced by the use of adequate volumes of water applied through spray or fog nozzles.

- 6.1.2. Water should be used on liquid ammonia spills only if sufficient water is available. For the purpose of this section, sufficient water may be taken to be 100 parts of water to one part of ammonia.
- 6.1.3. If an ammonia container is exposed to fire and cannot be removed, water should be used to cool it.
- 6.1.4. Under some circumstances ammonia in a container is colder than the available water supply. Under these circumstances water should not be sprayed on the container walls since it would heat the ammonia and aggravate any gas leak.
- 6.1.5. If it is found necessary to dispose of ammonia, as from a leaking container, liquid ammonia should be discharged into a vessel containing water sufficient to absorb it. Sufficient water may be taken to be ten parts of water per part ammonia. The ammonia should be injected into the water as near the bottom of the vessel as practical.

6.2. Safety

- 6.2.1. Personnel required to handle ammonia shall be trained in safe operating practices and the proper action to take in the event of emergencies. Personnel shall be instructed to use the equipment listed in 6.2.3 in the event of an emergency.
- 6.2.2. If a leak occurs in an ammonia system, the personnel trained for and designated to act in such emergencies shall:
 - 6.2.2.1. See that persons not required to deal with an emergency are evacuated from the contaminated area.
 - 6.2.2.2. Put on a suitable gas mask.
 - 6.2.2.3. Wear gauntlet type plastic or rubber gloves and wear plastic or rubber suits in heavily contaminated atmospheres.
 - 6.2.2.4. Shut off the appropriate valves.
- 6.2.3. All permanent storage facilities shall have on hand, as a minimum, the following equipment:
 - 6.2.3.1 One full face gas mask.
 - 6.2.3.2. One pair of protective gloves.
 - 6.2.3.3. One pair of protective boots.
 - 6.2.3.4. One protective slicker and/or protective pants and jacket.
 - 6.2.3.5. Easily accessible shower and/or at least 50 gallons of clean water in an open top container.
 - 6.2.3.6. Tight fitting vented goggles or one full face shield.
- 6.2.4. Where several persons are usually present, additional safety equipment may be desirable.

- 6.2.5. Each tank motor vehicle transporting anhydrous ammonia, except farm applicator vehicles, shall carry a container of at least five gallons of water and shall be equipped with at least one pair of protective gloves and a full face gas mask, a pair of tight-fitting goggles or one full face shield. The driver shall be instructed in their use and the proper action to take to provide for his safety.
- 6.2.6. If a leak occurs in transportation equipment and it is not practical to stop the leak, the driver should move the vehicle to an isolated location away from populated communities or heavily traveled highways.
- 6.2.7. If liquid ammonia contacts the skin or eyes, the affected area should be promptly and thoroughly flushed with water. Do not use neutralizing solutions or ointments on affected areas. A physician shall treat all cases of eye exposure to liquid ammonia.
- 6.2.7. Gloves, boots, slickers, jackets and pants shall be made of rubber or other material impervious to ammonia.

6.3. Filling Densities

- 6.3.1. The filling densities for containers shall not exceed the following:

	Aboveground	Underground
(1) Uninsulated	56%*	58%
(2) Insulated	57%	
(3) DOT containers shall be filled in accordance with DOT Regulation		

*This corresponds to 82% by volume at -28°F, 85% by volume at 5°F, 87.5% by volume at 30°F, and 90.6% by volume at 60°F.

- 6.3.2. If containers are to be filled according to liquid level by any gauging method other than a fixed length dip tube gauge, each container should have a thermometer well so that the internal liquid temperature can be easily determined and the amount of liquid and vapor in the container corrected to a 60°F basis.

6.4. Transfer of Liquids

- 6.4.1. Anhydrous ammonia shall always be at a temperature suitable for the material of construction and design of the receiving containers and shall comply with Appendix R of API Standard 620 "Recommended Rules for Design and Construction of Large Welded Low-Pressure Storage Tanks" for materials for low temperature service.
- 6.4.2. At least one attendant shall supervise the transfer of liquids from the time the connections are first made until they are finally disconnected.
- 6.4.3. Flammable gases or gases which will react with ammonia (such as air) shall not be used to unload tank cars or transport trucks.
- 6.4.4. Containers shall be charged or used only upon authorization of the owner.
- 6.4.5. Containers shall be gauged and charged only in the open atmosphere or in buildings approved for that purpose.
- 6.4.6. Pumps used for transferring ammonia shall be recommended and labeled for ammonia service by the manufacturer.
- 6.4.6.1. Pumps shall be designed for at least 250 psig working pressure.

- 6.4.6.2. Positive displacement pumps shall have installed, off the discharge port, a constant differential relief valve discharging into the suction port of the pump through a line of sufficient size to carry the full capacity of the pump at relief valve setting, which setting and installation shall be according to pump manufacturer's recommendations.
- 6.4.6.3. On the discharge side of the pump, before the relief valve line, there shall be installed a pressure gauge graduated from 0 to 400 psig.
- 6.4.6.4. Plant piping shall contain shut-off valves located as close as practical to pump connections.
- 6.4.7. Compressors used for transferring or refrigerating ammonia shall be recommended and labeled for ammonia service by the manufacturer.
 - 6.4.7.1. Compressors shall be designed for at least 250 psig working pressure. Crank cases of compressors not designed to withstand system pressure shall be protected with a suitable safety relief valve.
 - 6.4.7.2. Plant piping shall contain shut-off valves located as close as practical to compressor connections.
 - 6.4.7.3. A safety relief valve large enough to discharge the full capacity of the compressor shall be connected to the discharge before any shut-off valve.
 - 6.4.7.4. Compressors shall have pressure gauges at suction and discharge graduated to at least one and one-half times the maximum pressure that can be developed.
 - 6.4.7.5. Where necessary to minimize entry of liquid into the compressor, adequate means, such as drainable liquid trap, shall be provided on the suction side of the compressor.
 - 6.4.7.6. Where necessary to prevent contamination, an oil separator shall be provided on the discharge side of the compressor.
- 6.4.8. Loading and unloading systems shall be protected by suitable devices to prevent emptying of the storage container or the container being loaded or unloaded in the event of severance of the hose. Backflow check valves or properly sized excess flow valves shall be installed where necessary to provide such protection. In the event that such valves are not practical, remotely operated shut-off valves may be installed.
- 6.4.9. Meters used for the measurement of liquid anhydrous ammonia shall be recommended and labeled for ammonia service by the manufacturer and meet the requirements of The Measurement Standards Act, §§ 35-14-101 through 134, C.R.S.
 - 6.4.9.1. Liquid meters shall be designed for a minimum working pressure of 250 psig.
 - 6.4.9.2. The metering system shall incorporate devices that will prevent the inadvertent measurement of vapor.

6.5. Tank Car Unloading Points and Operations

- 6.5.1. Unloading operations shall be performed by reliable persons properly instructed and made responsible for careful compliance with all applicable procedures.

- 6.5.2. Caution signs shall be so placed on the track or car as to give necessary warning to persons approaching car from open end or ends of siding and shall be left up until after car is unloaded and disconnected from discharge connections. Signs shall be of metal or other suitable material, at least 12 by 15 inches in size and bear the words "STOP-Tank Car Connected" or "STOP-Men At Work" the word "STOP", being in letters at least 4 inches high and the other words in letters at least 2 inches high. The letters shall be white on a blue background.
- 6.5.3. The track of a tank car siding shall be substantially level.
- 6.5.4. Brakes shall be set and wheels blocked on all cars being unloaded.
- 6.5.5. Tank cars of anhydrous ammonia shall be unloaded only at approved locations meeting the requirements of Rules 6.2.3 and 6.4.8.

SECTION 7. SYSTEMS UTILIZING STATIONARY, PIER-MOUNTED OR SKID-MOUNTED ABOVEGROUND OR UNDERGROUND

This section applies to stationary, pier-mounted, skid-mounted, aboveground or underground storage installations utilizing containers other than those constructed in accordance with Department of Transportation Specifications. All Rules of Section 2 apply to this section unless otherwise noted.

7.1. Design Pressure and Construction of Containers

- 7.1.1. The minimum design pressure for non-refrigerated aboveground containers shall be 250 psig. [See.1.2.1.]

NOTE: U-68 and U-69 ASME Code containers with a design pressure of 200 psig are acceptable if recertified to 250 psig and equipped with safety relief valves set at 250 psig as permitted in 2.7.2.

7.2. Container Valves and Accessories, Filling and Discharge Connections

- 7.2.1. Each filling connection shall be provided with combination back-pressure check valve and excess flow valve; one double or two single back-pressure check valves; or a positive shut-off valve in conjunction with either an internal back-pressure check valve or an internal excess flow valve.
- 7.2.2. All vapor and liquid connections, except safety relief valves and those specifically exempt in Rules 2.4.5 and 2.4.6. shall be equipped with approved excess flow valves; or in lieu thereof, may be fitted with approved quick-closing internal valves which, except during operating periods, shall remain closed.
- 7.2.3. Each storage container shall be provided with a pressure gauge graduated from 0 to 400 psig. Gauges shall be designated for use in ammonia service.
- 7.2.4. All containers shall be equipped with an approved vapor return valve.
- 7.2.5. All containers shall be equipped with a fixed maximum liquid level gauge.

7.3. Safety Relief Devices

- 7.3.1. Every container shall be provided with one or more safety relief valves of spring-loaded or equivalent type and shall comply with the following:

- 7.3.1.1. The discharge from safety relief valves shall be directed away from the container upward and unobstructed to the open air. Vent pipes shall not be restrictive or smaller in size than the safety relief valve outlet connection. All safety relief valve discharges shall have suitable rain caps that will allow free discharge of the vapor and prevent the entrance of water. Suitable provision shall be made for draining condensate which may accumulate.
- 7.3.1.2. If desired, vent pipes from two or more safety relief devices located on the same unit, or similar lines from two or more different units, may be run into a common header, provided the cross-sectional area of such header is at least equal to the sum of the cross-sectional areas of the individual vent pipes.
- 7.3.2. The rate of discharge of spring-loaded safety relief valves installed on underground containers may be reduced to a minimum of 30 per cent of the rate of discharge specified in Appendix A of this Rule. Containers so protected shall not be uncovered after installation until the liquid ammonia has been removed. Containers which may contain liquid ammonia before being installed underground and before being completely covered with earth are considered aboveground containers when determining the rate of discharge requirements of the safety relief valves.
- 7.3.3. On underground installations where there is a probability of the manhole or housing becoming flooded, the discharge from vent lines shall be located above the high water level. All manholes or housings shall be provided with ventilated louvers or their equivalent, the area of such openings equaling or exceeding combined discharge areas of safety relief valves and vent lines which discharge their content into the manhole housing.
- 7.4. Installation of Storage Containers**
- 7.4.1. Containers installed aboveground shall be provided with substantial reinforced concrete footings and foundations or structural steel supports mounted on reinforced concrete foundations. In either case, the reinforced concrete foundations or footings shall extend below the established frost line and shall be of sufficient width and thickness to support the total weight of the containers and contents adequately. The foundation shall maintain the lowest point of the tank at not less than 18 inches above the ground. Floating type foundations shall also be acceptable providing the foundations are designed to adequately support the tank, contents and pumping equipment. Crushed rock of adequate depth would be acceptable.
- 7.4.2. Horizontal aboveground containers shall be mounted on foundations in such a manner as to permit expansion and contraction. Every container shall be supported so as to prevent the concentration of excessive loads on the supporting portion of the shell. The bearing afforded by the saddles shall extend over at least one third of the circumference of the shell. Suitable means for preventing corrosion shall be provided on that portion of the container in contact with the foundations or saddles.
- 7.4.3. Containers buried underground shall be placed so that the top of the container is at least one foot below the surface of the ground. Should ground conditions make compliance with these requirements impracticable, precautions shall be taken to prevent physical damage to the container. It is not necessary to cover the portion of the container to which a manhole and other connections are affixed. When necessary to prevent floating, containers shall be securely anchored or weighted.

- 7.4.4. Underground containers shall be set on firm foundations (firm earth may be used) and surrounded with soft earth or sand well tamped in place. As a further means of resisting corrosion, the container, prior to being placed underground, shall be given a protective coating satisfactory to the Commissioner. Such protective coating shall be equivalent to hot dip galvanizing, or to two preliminary coatings of red lead followed by a heavy coating of coal tar or asphalt. The container thus coated shall be lowered into place in such a manner as to prevent abrasion or other damage to the coating.
- 7.4.5. Distance between aboveground and underground containers of over 1,200 gallons capacity shall be at least five feet
- 7.4.6. Secure anchorage or adequate pier height shall be provided against container flotation wherever sufficiently high flood water might occur.

7.5. Reinstallation of Containers

- 7.5.1. Containers once installed underground shall not later be reinstalled aboveground or underground, unless they successfully withstand hydrostatic pressure retests at the pressure specified for the original hydrostatic test as required by the code under which the tank was constructed and show no evidence of serious corrosion.
- 7.5.2. Where containers are reinstalled underground the corrosion resistant coating shall be put in good condition; see 7.4.4. Where containers are reinstalled aboveground, safety relief devices or gauging devices shall comply with 2.7 and 7.3 respectively for aboveground containers.

7.6. Marking of Containers

- 7.6.1. Each container or group of containers shall be marked on at least two sides with the words "Anhydrous Ammonia" in sharply contrasting colors with letters not less than four inches high and displayed a National Fire Protection Association diamond for anhydrous ammonia in a location that is readily visible to emergency responders.

7.7. Protection of Container Appurtenances

- 7.7.1. Valves and other appurtenances shall be protected against physical damage. Main container shut-off valves shall be kept closed and locked when the installation is unattended. If the facility is protected against tampering by fencing or other suitable means, valve locks are not required.
- 7.7.2. All connections to underground containers should be located within a substantial dome, housing or manhole fitted with a substantial removable cover. Appurtenances shall also be protected during the transit of containers intended for installation underground.
- 7.7.3. Storage containers need not be grounded.

7.8. Identification

- 7.8.1. A sign shall be displayed in a conspicuous place stating the name, address, and phone number of the nearest representative, agent, or owner of the storage system.

SECTION 8. SYSTEMS UTILIZING PORTABLE DOT CONTAINERS

This Section 8 incorporates by reference the specification of the U.S. Department of Transportation systems utilizing cylinders, portable tanks (DOT-51) 49 CFR 173.32, or "ton containers" (DOT-106A, DOT-110A 49 CFR 179.300-179.301), dated October 1, 2011. All General Rules of Section 2 apply to this Section 8, unless otherwise noted.

8.1. Containers

- 8.1.1. Containers shall comply with Department of Transportation Specifications and shall be maintained, filled, packaged, marked, labeled and shipped to comply with current DOT Regulations 49 CFR Parts 171-177 and compressed Gas Association Guide to Classification and Labeling of Compressed Gases, CGA C-7 2014.
- 8.1.2. Containers shall be stored in an area free from ignitable debris and in such manner as to prevent external corrosion. Storage may be indoors or outdoors.
- 8.1.3. Containers shall not be buried below ground.
- 8.1.4. Containers shall be set upon firm foundations or otherwise firmly secured. The possible effect of sending on the outlet piping shall be guarded against by a flexible connection or special framing.
- 8.1.5. Containers shall be protected from heat sources such as radiant flame and steam pipes. Do not apply heat directly to containers to raise the pressure.
- 8.1.6. Containers shall be stored in such manner as to protect them from moving vehicles or external damage.
- 8.1.7. Any container which is designed to have a valve protection cap shall have the cap securely in place when the container is not in service.

8.2. Container Valves and Regulating Equipment

- 8.2.1. Container valves and pressure regulating equipment shall be protected against tampering when installed for use.
- 8.2.2. Container valves shall be protected while in transit, in storage, and while being moved into final utilizations, as follows:
 - 8.2.2.1. By setting them into a recess of the container, or
 - 8.2.2.2. By ventilated cap or collar, fastened to the container, capable of withstanding a blow from any direction equivalent to that of a 30-lb. weight dropped four feet. Construction must be such that a blow will not be transmitted to the valves or other connections.
- 8.2.3. When containers are not connected for service, the outlet valves shall be kept tightly closed even though containers are considered empty.

8.3. Safety Relief Devices

- 8.3.1. Containers shall be provided with safety relief devices as required by Department of Transportation Regulations 49 CFR Part 173.315(i), October 1, 2011.

SECTION 9. SYSTEMS MOUNTED ON TRUCKS, SEMI-TRAILERS, AND TRAILERS FOR TRANSPORTATION OF AMMONIA

This Section applies specifically to systems mounted on trucks, semi-trailers and trailers (other than those covered under Sections 10 and 11) used for the transportation of ammonia. All General Rules of Section 2 apply to this Section 9 unless otherwise noted.

9.1. Design Pressure of Containers

- 9.1.1. Containers shall be constructed in accordance with Rule 3.1 with a minimum design pressure of 250 psig.
- 9.1.2. The shell or head thickness of any container shall not be less than 3/16 inch.
- 9.1.3. All container openings, except safety relief valves, liquid level gauging devices and pressure gauges, shall be labeled to designate whether they communicate with liquid or vapor space. Labels may be on valves.
- 9.1.4. Baffles are not required for cargo tanks.

9.2. Mounting Containers on Truck

- 9.2.1. The means of attachment of any container to the cradle, frame or chassis of a vehicle shall be designed on a basis of two "g" loading in either direction, using a safety factor of not less than 4, based on the ultimate strength of the material used. For purposes of this requirement, two "g" of load support is equivalent to three times the static weight of the articles supported; two "g" of loading and bending, acceleration, and torsion is equivalent to twice the static weight support applied horizontally at the road surface.
- 9.2.2. "Hold-down" devices, when used, shall anchor the container to the cradle, frame or chassis in a suitable and safe manner that will not introduce undue concentration of stresses. These devices shall incorporate positive means for drawing the container down tight, and suitable stops or anchors shall be provided to prevent relative movement between container and framing due to stopping, starting or changes in direction.
- 9.2.3. Vehicles designed and constructed so that the cargo tanks constitute in whole or in part the stress member used in lieu of the frame shall be supported by external cradles suspending at least 120° of the shell circumference. The design calculation shall include beam stress, shear stress, torsion stress, bending moment and acceleration stress, in addition to those covered by the code under which the cargo tank was designed.
- 9.2.4. If a liquid withdrawal line is installed in the bottom of a container, the connections thereto, including hose, shall not be lower than the lowest horizontal edge of the trailer axle.
- 9.2.5. Provisions shall be made to secure both ends of the hose while in transit.
- 9.2.6. When the cradle and the container are not welded together, suitable material shall be used between them to eliminate metal-to-metal friction.

9.3. Container Appurtenances

- 9.3.1. Non-recessed container fittings and appurtenances shall be protected against physical damage by either: (1) a protected location, (2) the vehicle frame or bumper, or (3) a protective housing. The protective housing, if used, shall comply with the requirements under which the containers are fabricated with respect to design and construction, and shall be designed to withstand static loadings in any direction equal to twice the weight of the container and attachments when filled with the lading using a safety factor of not less than 4, based on the ultimate strength of the material to be used. The protective housing if used shall be protected with a weather cover, if necessary, to ensure proper operation of valves and safety relief devices.

- 9.3.2. All connections to containers, except filling connections (see 9.3.3), safety relief devices, and liquid level and pressure gauge connections, shall be provided with suitable automatic excess flow valves, or in lieu thereof, may be fitted with quick-closing internal valves, which shall remain closed except during delivery operations. The control mechanism for such valves may be provided with a secondary control remote from the delivery connections and such control mechanism shall be provided with a fusible section (melting point 208°F to 220°F) which will permit the internal valve to close automatically in case of fire.
- 9.3.3. Filling connections shall be provided with automatic back-pressure check valves, excess-flow check valves, or quick-closing internal valves, to prevent back-flow in case the filling connection is broken. Where the filling and discharge connect to a common opening in the container shell and that opening is fitted with a quick-closing internal valve as specified in 9.3.2, the automatic valve shall not be required.
- 9.3.4. All containers shall be equipped for spray loading (filling in the vapor space) or with an approved vapor return valve of adequate capacity.
- 9.3.5. All containers shall be equipped with a fixed maximum liquid level gauge.
- 9.3.6. All containers shall be equipped with a pressure-indicating gauge having a dial graduated from 0-400 psig.

9.4. Piping and Fittings

- 9.4.1. All piping, tubing and fittings shall be securely mounted and protected against physical damage.
- 9.4.2. Piping used on non-refrigerated systems shall be at least ASTM A-53 Grade B Electric Resistance Welded and Electric Flash Welded Pipe or equal. Such pipe shall be at least Schedule 40 when joints are welded, or welded and flanged. Such pipe shall be at least Schedule 80 when joints are threaded. Brass, copper, or galvanized steel pipe or tubing shall not be used.
- 9.4.3. The truck unloading line shall be provided with an excess flow valve at the hose connection unless an approved quick dosing internal valve is provided in the container unloading connection. (See 9.3.2)

9.5. Safety Relief Devices

- 9.5.1. The discharge from container safety relief valves shall be vented away from the container upward and unobstructed to the open air in such a manner as to prevent any impingement of escaping gas upon the container; loose fitting rain caps shall be used. Size of discharge lines from safety relief valves shall not be smaller than the nominal size of the safety relief valve outlet connection. Suitable provision shall be made for draining condensate which may accumulate in the discharge pipe.

9.6. Marking of Container

- 9.6.1. Every container, whether loaded or empty, shall be conspicuously and legibly marked on each side and rear thereof on a background of sharply contrasting color with the words "COMPRESSED GAS" in letters at least four inches high; and with the words "ANHYDROUS AMMONIA" in letters at least four inches high.

9.7. Transfer of Liquids

- 9.7.1. The content of tank motor vehicle containers shall be determined by weight, or suitable metering device.

NOTE: If the content of a container is to be determined by liquid level measurement, the container shall have a thermometer well so that the internal liquid temperature can be easily determined. This volume when converted to weight shall not exceed the filling density.

- 9.7.2. Pumps or compressors shall be designed and installed in accordance with Rule 6.4 and protected against physical damage when mounted upon ammonia tank trucks and trailers.
- 9.7.3. Tank motor vehicles of greater than 3500 water gallons capacity shall be unloaded only at approved locations meeting the requirements of Rules 6.2.3 and 6.4.8.

9.8. Trailers and Semi-Trailers

- 9.8.1. Trailers shall be firmly and securely attached to the vehicle drawing them by means of suitable drawbars, supplemented by suitable safety chain (or chains) or safety cables.
- 9.8.2. Every trailer and semi-trailer shall be equipped with an emergency braking system to be activated in the event of hitch failure.
- 9.8.3. Trailers shall be of a type of construction which will prevent the towed vehicle from whipping or swerving dangerously from side to side and which will cause it to follow substantially in the path of the towing vehicle.
- 9.8.4. Where a fifth wheel is employed on a semi-trailer, it shall be ruggedly designed, securely fastened to both units, and equipped with a positive locking mechanism which will prevent separation of the two units except by manual release.
- 9.8.5. Every trailer or semi-trailer shall be provided with side lights and a tail light.

9.9. Electrical Equipment and Lighting

- 9.9.1. Tank trucks, tank trailers, and tank semi-trailers, may not be equipped with any artificial light other than electric light. Electric lighting circuits shall have suitable overcurrent protection (fuses or automatic circuit breakers). The wiring shall have sufficient carrying capacity and mechanical strength, and shall be suitably secured, insulated and protected against physical damage.

9.10. Protection Against Collision

- 9.10.1. Each tank motor vehicle shall be provided with properly attached bumpers or chassis extensions arranged to protect the tank, piping, valves and fittings from physical damage in case of collision.

9.11. Chock Blocks

- 9.11.1. At least two chock blocks shall be provided. These blocks shall be placed to prevent rolling of the vehicle whenever it is parked during loading and unloading operations.

9.12. Portable Tanks (Including Skid Tanks)

- 9.12.1. When portable tanks are used in lieu of cargo tanks and are permanently mounted on tank motor vehicles for the transportation of ammonia, they shall comply with the requirements of Section 9. Where portable tanks, including those built to DOT Specification 51, 106A or 110A, are used for farm storage they shall comply with Section 7. When portable tanks are used as shipping containers in interstate commerce they shall comply with Section 8.

9.13. Safety Equipment

9.13.1. All tank trucks, trailers, and semitrailers shall be equipped with the following:

- 9.13.1.1. One full face gas mask with anhydrous ammonia refill canisters.
- 9.13.1.2. One pair of protective gloves made of rubber or other material impervious to ammonia.
- 9.13.1.3. Tight-fitting goggles or one full face shield.
- 9.13.1.4. A container of not less than five gallons of readily available clean water.

*An ammonia canister is effective for short periods of time in light concentrations of ammonia vapor, generally 15 minutes in concentrations of 3% and will not protect breathing in heavier concentrations. If ammonia vapors are detected when mask is applied the concentration is too high for safety. The life of a canister in service is controlled by the percentage of vapors to which it is exposed. Canisters must not be opened until ready for use and should be discarded after use. Unopened canisters may be guaranteed for as long as three years. All should be dated when received because of this limited life. In addition to this protection, an independently supplied air mask of the type used by fire departments may be used for severe emergencies.

SECTION 10. SYSTEMS MOUNTED ON FARM WAGONS (IMPLEMENTS OF HUSBANDRY) FOR THE TRANSPORTATION OF AMMONIA

This Section applies to containers of 3000 gallons capacity or less and pertinent equipment mounted on farm wagons (implements of husbandry) and used for the transportation of ammonia. All General Rules of Section 2 apply to this Section unless otherwise noted.

10.1. Design of Containers

10.1.1. Containers shall be constructed in accordance with Rule 3.1.

10.2. Mounting Containers

- 10.2.1. A suitable "stop" or "stops" shall be mounted on the farm wagon or on the container in such a way that the container shall not be dislodged from its mounting due to farm wagon coming to a sudden stop. Back slippage shall also be prevented by proper methods.
- 10.2.2. A suitable "hold-down" device shall be provided which will anchor the container to the farm wagon at one or more places on each side of the container.
- 10.2.3. When containers are mounted on four-wheel farm wagons, care shall be taken to ensure that the weight is distributed evenly over both axles.
- 10.2.4. When the cradle and the container are not welded together, suitable material shall be used between them to eliminate metal-to-metal friction.

10.3. Container Appurtenances

- 10.3.1. All containers shall be equipped with a fixed maximum liquid level gauge.
- 10.3.2. All containers with a capacity exceeding 250 gallons shall be equipped with a pressure gauge having a dial graduated from 9-400 psi.
- 10.3.3. The filling connection shall be fitted with combination back-pressure check valve and excess-flow valve; one double or two single back-pressure check valves; or a positive shut-off valve in conjunction with either an internal back-pressure check valve or an internal excess flow valve.

- 10.3.4. All containers with a capacity exceeding 250 gallons shall be equipped for spray loading or with an approved vapor return valve.
- 10.3.5. All vapor and liquid connections, except safety relief valves and those specifically exempt in 2.4.5.6, shall be equipped with approved excess flow valves or may be fitted with quick-closing internal valves which, except during operating periods, shall remain closed.
- 10.3.6. Fittings shall be protected from physical damage by means of a rigid guard designed to withstand static loading in any direction equal to twice the weight of the container and lading using a safety factor of four (4) based upon the ultimate strength of the material used. If the guard is fully enclosed, the safety relief valves shall be properly vented through the guard.
- 10.3.7. If a liquid withdrawal line is installed in the bottom of a container, the connections thereto, including hose, shall not be lower than the lowest horizontal edge of the farm wagon axle.
- 10.3.8. Provision shall be made to secure both ends of the hose while in transit.

10.4. Marking of Container

- 10.4.1. There shall appear on each side and on the rear end of the container in letters at least four inches high, the words "ANHYDROUS AMMONIA".

10.5. Farm Wagons (Implements of Husbandry)

- 10.5.1. Farm wagons (Implements of Husbandry) shall conform with State Regulations.
- 10.5.2. All farm wagons shall be securely attached to the vehicle drawing them by means of drawbars supplemented by suitable safety chains.
- 10.5.3. A farm wagon shall be constructed so that it will follow substantially in the path of the towing vehicle and will prevent the towed farm wagon from whipping or swerving dangerously from side to side.
- 10.5.4. All farm wagons shall have five (5) gallons or more of readily available clean water, one pair of tight fitting goggles or full face shield, and one pair of protective gloves.

SECTION 11. SYSTEMS MOUNTED ON FARM EQUIPMENT (IMPLEMENTS OF HUSBANDRY) FOR THE APPLICATION OF AMMONIA

This Section applies to systems mounted on farm equipment and used for the field application of ammonia. All General Rules of Section 2 apply to this Section unless otherwise noted.

11.1. Design of Containers

- 11.1.1. The minimum design for containers shall be in accordance with Rule 3.1.

11.2. Mounting of Containers

- 11.2.1. All containers shall be securely mounted.

11.3. Container Valves and Appurtenances

- 11.3.1. Each container shall have a fixed maximum liquid level gauge.

- 11.3.2. The filling connection shall be fitted with combination back-pressure check valve and excess-flow valve; one double or two single back-pressure check valves; or a positive shut-off valve in conjunction with either an internal back-pressure check valve or an internal excess-flow valve.
- 11.3.3. An excess-flow valve is not required in the vapor connection, provided the controlling orifice is not in excess of seven sixteenths (7/16) of an inch in diameter and the valve is a hand-operated (attached hand wheel or equivalent) shut-off valve. To assist in filling applicator tanks, it is permissible to bleed vapors to the open air, providing the preceding requirements are met.
- 11.3.4. Metering devices may be connected directly to the tank withdrawal valve. A union type connection is permissible between the tank valve and metering device. Remote mounting of metering devices is permissible using hose which meets with specifications set out in Appendix B of this Rule. When the applicator tank is trailed and the metering device is remotely mounted, such as on the tractor tool bar, an automatic break-a-way type, self-closing, coupling must be used.
- 11.3.5. No excess-flow valve is required in the liquid withdrawal line provided the controlling orifice between the contents of the container and the outlet of the shut-off valve (see 2.5.5) does not exceed 7/16 inch in diameter.

SECTION 12. ADDITIONAL SAFETY REQUIREMENTS

- 12.1. The following minimum requirements of equipment shall be on hand at all places wherever Anhydrous Ammonia is handled or transported, if not otherwise prescribed in these Rules:
 - 12.1.1. One pair of tight fitting goggles or full face shield.
 - 12.1.2. One pair of protective gloves.
 - 12.1.3. A container of not less than five gallons of readily available clean water.
- 12.2. It shall be considered a violation if, at any time in the handling of Anhydrous Ammonia, the prescribed safety equipment is not worn or used.
- 12.3. Agricultural anhydrous ammonia tank cars and transport trucks shall be unloaded only through permanent, approved unloading sites and into permanently installed bulk storage tanks, except as otherwise provided below.
 - 12.3.1. Agricultural anhydrous ammonia may be unloaded directly from a railroad tank car into a U.S. Department of Transportation approved over-the-road transport truck of more than 3500 gallons capacity only for the purpose of transport to bulk storage tanks which do not have access to a railroad siding.
 - 12.3.2. Tank cars and transport trucks may be unloaded at permanent, approved unloading sites into portable acid-fertilizer conversion units for the purpose of producing liquid fertilizer, if such conversion units have approved, automatic, normally closed valves in the line connecting the source tank and the conversion unit.

SECTION 13. REGISTRATION

On or before January 1 of each year, every person who owns one or more anhydrous ammonia storage tanks, mobile transportation tanks, or tank-mounted applicators within this state shall register each of such tanks or applicators with the Department and shall pay a registration fee as established by the Agricultural Commission.

SECTION 14. STATEMENTS OF BASIS, SPECIFIC STATUTORY AUTHORITY AND PURPOSE

The Statement of Basis, Specific Statutory Authority and Purpose for rulemaking activity from 1992 is no longer in the Department's files.

1.4.1. October 9, 2008 – Effective November 30, 2008

STATUTORY AUTHORITY:

The Commissioner of Agriculture, Colorado Department of Agriculture, adopts these permanent rules pursuant to the provisions and requirements of the Anhydrous Ammonia Act, 35-13-101 - 109, C.R.S.

PURPOSE:

The purpose of these rules is to:

Implement the statutory change to the Anhydrous Ammonia Act which require the Commissioner to establish rules to specify the dates for the registration of anhydrous ammonia storage tanks, mobile transportation tanks and tank mounted applicators.

Correct typographical errors.

FACTUAL POLICY AND ISSUES;

The factual and policy issues encountered in the proposal of these permanent rules is as follows:

On August 6, 2008, The Anhydrous Ammonia Act was amended by Senate Bill 08-097. Section 35-13-109 (1) requires the Commissioner to specify by rule the dates for the registration of anhydrous ammonia storage tanks, mobile transportation tanks and tank mounted applicators.

14.2. Adopted November 10, 2015- Effective December 30, 2015

STATUTORY AUTHORITY:

The Commissioner's authority for the adoption of these Rule amendments is set forth in Section 35-13-103, C.R.S.

Purpose:

The purpose of these Rule Amendments is to:

1. Delete obsolete language at the beginning of the Rules;
2. Correct typographical errors throughout the rules;
3. Amend Rules by deleting language that is not essential and unenforceable including reference materials;
4. Amend Rules to exempt refrigerated ammonia systems and delete all refrigerated ammonia rules;
5. Delete references to "Appendix C" within the Rules;
6. Add Rule 1.3. to incorporate by reference all of the standards referred to in the rules, instruct where to inspect codes, and to explain later amendments of incorporated material are not included;

7. Amend Rules by changing the language “this Standard” to “these Rules” to be consistent with other Colorado Department of Agriculture rules;
8. Amend Rules by changing the word “paragraph” with the word “Rule” throughout the rules so as to be consistent with other Colorado Department of Agriculture rules;
9. Replace the language “Authority having Jurisdiction” with the word “Commissioner” throughout the rules;
10. Amend Rules to add definitions necessary to understand the Rules.
11. Amend Rule 1.4.8. by adding the current edition of the unfired pressure vessel code and striking obsolete references to The Joint Code of the American Petroleum Institute and The American Society of Mechanical Engineers (API-ASME Code) 1951 edition;
12. Delete the definition of “gas mask” in Rule 1.3.11.;
13. Add Rule 3.1.5. to include a records retention requirement of inspections and welding on the container;
14. Realign the order of the Rules to be consistent with the requirements listed in Statute for the design, construction, location, installation, and operation of anhydrous ammonia systems;
15. Amend Rule 6.4.9. by replacing the words “state Weights & Measures” with “The Measurement standards Act, §§ 35-14-101 through 134, C.R.S.”;
16. Amend Rules 6.2.3. and 9.13.1. by replacing language referring to emergency rescue with personal protection equipment.
17. Amend Rules to include the specific code of federal regulations cites throughout the Rules.

Factual and Policy Basis:

The factual and policy issues pertaining to the adoption of these Rule amendment are as follows:

1. The current Rules have not been revised for a very long time and therefore these Rules contain antiquated language and formatting requirements. These Rule changes are being made to modernize the language where appropriate, comply with the current Rule format requirements, and to align the Rule with the Statute;
2. Obsolete language at the beginning of the Rule has been deleted to comply with the Department’s current Rule format;
3. References to refrigerated anhydrous systems have been deleted throughout the Rule to clarify that these Rules only regulate anhydrous ammonia used as an agricultural fertilizer. The amended Rules are 1.1.1., 1.1.2.4., Section 4, 6.3.1., 6.4.1., and 6.4.7.1.;
4. Rules referring to Appendix C have been deleted because Appendix C does not exist;
5. Definitions have been added to clarify the Rule. The amended Rules are 1.4.3.(ASME), 1.4.4.(Backflow check valve), 1.4.6.(Chemical-splash goggles), 1.4.10.(Data plate), 1.4.11.(Densely populated area), 1.4.13.(Emergency shutoff valve), 1.4.14.(Excess-flow valve), 1.4.19.(Mobil container), 1.4.20.(Non-code welding), 1.4.21.(Permanent storage container), 1.4.22.(Permanent storage facility), and 1.4.23.(Place of public assembly);

6. References to materials incorporated by reference within the Rule have been updated. The amended Rules are 1.3.1., 1.3.2., and 1.3.3.;
7. Informational language not essential to the Rule has been deleted. The deleted language is not necessary for the administration of the Rule. The amended Rules are 1.1.2.2., 1.2.7., 1.2.8., 1.2.9., and 1.4.7.;
8. Language not essential to the Rule has been deleted or replaced to clarify the intent of the Rule. The amended Rules are 1.1.1., 1.2.1., 1.4., 1.4.1., 1.4.15., 9.7.1., 9.7.3., 10.1.1., 11.1.1., and 12.1.;
9. References to other resources not essential to the enforcement of the Rule were deleted to avoid confusion as to which references are needed to be in compliance with the Rule. The amended Rules are 1.4.3., 1.4.8., and after Rule 6.2.;
10. The wording “state Weights and Measures” in rule 6.4.9. was replaced with “The Measurement standards Act, §§ 35-14-101 through 134, C.R.S” to clarify where the requirements for meters used for the measurement of liquid anhydrous ammonia can be found.
11. References to emergency and rescue equipment were deleted in the Rule to eliminate the perception that untrained personnel should perform emergency and rescue operations. Emergency and rescue operations should be performed by trained professionals. The amended Rules are 1.3.11., 1.4.13., 6.2.3., and 9.13.1.;
12. The Rule was realigned to be consistent with the requirements of the Statute. The realignment will make finding the rules pertaining to the design, construction, location, installation, and operation of anhydrous ammonia systems for storing, handling, transporting, and utilizing anhydrous ammonia as an agricultural fertilizer within the Rule easier to find;
13. Records requirements for repairs completed on a container have been added to the Rule. The requirement will help the owner of the tank avoid costly re-examination and testing of the repaired tank before the tank can be registered and filled. The amended Rules are 1.4.15., and 3.1.5.;
14. The Rule was amended to clarify compliance with the Department of Transportation is required for all tanks transported by tank or trailer by adding the words “which shall be”, by adding the specific federal code site, and eliminating language that could cause confusion. Rule 1.1.1. further clarifies compliance with applicable Federal Statutes or Regulations. The amended Sections are 8. and 9. The amended Rules are 1.1.1., 8.1.1., 9.1.1., 9.6.1., and 10.4.1.
15. These revisions incorporate changes as a result of the Departments Regulatory Efficiency Review Process.

14.3. Adopted October 14, 2020 – Effective December 15, 2020

Statutory Authority

The Commissioner's authority for the adoption of this permanent Rule amendment is set forth in § 35-13-103 C.R.S.

Purpose

The purpose of this rulemaking is to update the language that incorporates regulations and standards by reference so as to comply with the new requirements in §24-4-103(12.5), C.R.S.

Factual and Policy Basis

This rulemaking updates the language and citations within the rule to properly incorporate regulations and standards by reference. More specifically, it provides the proper address for the Colorado Department of Agriculture, which changed in 2018, and provides the addresses of the agencies and organizations issuing the incorporated regulations and standards. This rulemaking also updates the edition for each incorporation.

APPENDIX A

Minimum required rate of discharge in cubic feet per minute of air at 120 per cent of the maximum permitted start to discharge pressure for safety relief valves to be used on containers other than those constructed in accordance with United States Department of Transportation cylinder specifications.

Surface Area Sq. Ft.	Flow Rate CFM Air	Surface Area Sq. Ft.	Flow Rate CFM Air	Surface Area Sq. Ft.	Flow Rate CFM Air	Surface Area Sq. Ft.	Flow Rate CFM Air
20	258	145	1,310	340	2,640	1,350	8,160
25	310	150	1,350	350	2,700	1,400	8,410
30	360	155	1,390	360	2,760	1,450	8,650
35	408	160	1,420	370	2,830	1,500	8,900
40	455	165	1,460	380	2,890	1,550	9,140
45	501	170	1,500	390	2,960	1,600	9,380
50	547	175	1,530	400	3,010	1,650	9,620
55	591	180	1,570	450	3,320	1,700	9,860
60	638	185	1,600	500	3,620	1,750	10,090
65	678	190	1,640	550	3,910	1,800	10,330
70	720	195	1,670	600	4,200	1,850	10,560
75	762	200	1,710	650	4,480	1,900	10,800
80	804	210	1,780	700	4,760	1,950	11,030
85	845	220	1,850	750	5,040	2,000	11,260
90	885	230	1,920	800	5,300	2,050	11,490
95	925	240	1,980	850	5,590	2,100	11,720
100	965	250	2,050	900	5,850	2,150	11,950
105	1,010	260	2,120	950	6,120	2,200	12,180
110	1,050	270	2,180	1,000	6,380	2,250	12,400
115	1,090	280	2,250	1,050	6,640	2,300	12,630
120	1,120	290	2,320	1,100	6,900	2,350	12,850
125	1,160	300	2,380	1,150	7,160	2,400	13,080
130	1,200	310	2,450	1,200	7,410	2,450	13,300
135	1,240	320	2,510	1,250	7,660	2,500	13,520
140	1,280	330	2,570	1,300	7,910		

Surface Area = Total Outside Surface Area of Container in Square Feet. When the Surface Area is not stamped on the name plate or when the marking is not legible, the area can be calculated by using one of the following formulas:

- (1) Cylindrical container with hemispherical heads

Area = overall length in feet times outside diameter in feet times 3.1416.

- (2) Cylindrical container with other than hemispherical heads

Area = (overall length in feet plus 0.3 outside diameter in feet) times outside diameter in feet times 3.1416.

- (3) Spherical container

Area = outside diameter in feet squared times 3.1416.

Flow Rate - CFM Air = cubic feet per minute of air required at standard conditions, 60°F and atmospheric pressure (14.7 psia).

The rate of discharge may be interpolated for intermediate values of surface area. For containers with total outside surface area greater than 2.500 sq. ft., the required flow rate can be calculated using the formula, Flow Rate CFM Air = $22.11 A^{0.82}$ where A = outside surface area of the container in square feet.

**APPENDIX B TFI-RMA SPECIFICATION FOR ANHYDROUS AMMONIA HOSE TFI-RMA
STANDARD NO. M-5**

1. SCOPE

This specification covers hose and hose assemblies commonly referred to as "pressure transfer hose," used to convey anhydrous ammonia liquid or to convey anhydrous ammonia gas where the gas is in contact with liquid ammonia. This specification primarily covers hose and hose assemblies which have a minimum burst pressure of 1750 psig, a safety factor of 5, and a maximum working pressure of 350 psig. These figures should not be misconstrued to mean that they are the maximum pressures to which anhydrous ammonia hose and hose assemblies are built, since higher pressure hose and hose assemblies are available for special applications.

2. SIZES AND TOLERANCES

Anhydrous ammonia hose shall be made with the following dimensions and tolerances:

RUBBER COVERED HOSE FOR USE WITH TWO-PIECE SCREW TYPE COUPLINGS			
I.D.	Tolerance	O.D.	Tolerance
1/2"	± 1/32"	15/16"	± 1/32"
3/4"	± 1/32"	1 1/4"	± 1/32"
1"	± 1/16"	1 1/2"	± 1/16"

NON-RUBBER COVERED AND RUBBER COVERED HOSE FOR USE WITH FULL FLOW COUPLINGS				
I.D.	Tolerance	O.D.	Tolerance	Nominal Tubing O.D.
13/32"	+ .039", - .015"	49/64"	± .031"	1/2"
1/2"	+ .047", - .015"	59/64"	± .031"	5/8"
5/8"	+ .047", - .015"	1-5/64"	± .031"	3/4"
7/8"	+ .047", - .015"	1-15/64"	± .031"	1"
1 1/8"	+ .062", - .015"	1 1/2"	± .047"	1 1/4"
1 3/8"	+ .062", - .015"	1 3/4"	± .047"	1 1/2"
1-13/16"	+ .062", - .015"	2-7/32"	± .047"	2"

HOSE FOR USE WITH OTHER TYPES OF COUPLINGS*	
I.D.	Tolerance
1/2"	± 1/32"
3/4"	± 1/32"
1"	± 1/16"
1 1/4"	± 1/16"
1 1/2"	± 1/16"
2"	± 1/16"

* The O.D. dimension and tolerance were intentionally omitted from this tabulation to provide for developments in both hose and couplings.

3. CONSTRUCTION

3.1 Inner Tube

The tube shall be uniform in quality and thickness and free from injurious defects. It shall meet the physical requirements of Section 4. The material shall be resistant to hardening or other deterioration due to the action of ammonia.

3.2 Reinforcement

The reinforcement shall consist of any material not adversely affected by permeating ammonia. The reinforcement shall be applied evenly and uniformly, and in such a way that it will meet the physical requirements of Section 4. In constructions utilizing a ply or plies of wire reinforcement, the composition of the wire shall be a suitable corrosion resistant stainless steel.

3.3 Cover

A rubber cover if used shall be uniform in quality and thickness and free from injurious defects. It shall meet the physical requirements of Section 4. The cover shall be so compounded or constructed that it will not blister in service, and will be resistant to deterioration due to the action of ammonia. A gas tight cover shall be pricked to relieve pressure build-up between inner tube and cover. The cover shall be resistant to deterioration due to exposure to the elements.

4. PHYSICAL TESTS

4.1 Tension Test of Tube and Cover

	Tube	Cover
Tensile, psi. min.	800	1200
Elongation, percent, min.	150	200

4.2 Adhesion Test

	Tube	Ply	Cover
Adhesion lbs./in.	10	8	10

- 4.2.1 In constructions having braided wire or woven wire filler reinforcing members, only the cover adhesion requirement will apply, as it is impractical to prepare adhesion test specimens except for determining cover adhesion.

4.3 Burst Test

All sizes have a minimum burst of 1750 psig. (See Scope).

4.4 Ammonia Performance Test

During the conditioning and flexing described in Sections 7.4 and 7.4.2 there shall be no evidence of cover blistering or leakage. At the conclusion of the conditioning and at the conclusion of the flexing test, the burst must still meet the requirements of Section 4.3. There shall be no evidence of separation of the component parts when the remainder of the samples are examined.

4.5 Low Temperature Test

The hose shall not fail at minus 40°F plus or minus 2°, when tested as described in Section 7.5, of this Specification.

5. TYPES OF TESTS

5.1 Acceptance Inspection

This includes all the tests specified, with the exception of the ammonia performance test.

5.2 Qualification Tests

The qualification tests are intended to establish that the hose is properly designed and constructed to give satisfactory service life. These tests shall be conducted by a recognized independent laboratory. The qualification tests shall consist of all the tests specified herein including the ammonia performance test.

6. METHOD OF SAMPLING

6.1 Acceptance Inspection

A 24-inch sample of each size and type hose, representative of the lot, shall be selected from each lot manufactured at one time, or from each 25,000 feet, whichever is smaller.

6.2 Qualification Test

In addition to the samples specified in Section 6.1, two 12-foot lengths of each size hose shall be selected for the ammonia performance test. Each new hose shall be subjected to a qualification test, and again whenever there has been a design change.

7. METHODS OF TESTING

7.1 Tension Test of Tube and Rubber Cover

The tension test shall be made in accordance with ASTM D-380.

7.2 Friction Test

The friction test shall be made in accordance with ASTM D-380.

7.3 Burst Test

The burst test shall be made in accordance with ASTM D-380 using the method entitled "Straight Bursting Test"

7.4 Ammonia Performance Test

Two 12-foot lengths of hose, to be marked "A" and "B", shall be filled with liquid anhydrous ammonia by connecting to a tank and flushing out with ammonia to remove all the air. One end of each length shall be sealed and the other end left connected to the liquid space of a tank of anhydrous ammonia. The hose shall then be conditioned for 14 days at ambient temperature of 60° to 100°F. A valve between the ammonia tank and the hose may be closed providing it is opened at least once each day to completely fill the hose with liquid anhydrous ammonia. The hose shall be examined each day for visible defects. There shall be no evidence of the cover blistering or perceptible leakage. If the hose is valved off at each end when liquid full, a hydrostatic relief valve should be provided between the block valves.

7.4.1 Conditioned Hose Burst Test

A 24-inch sample cut from hose marked "A" shall be subjected to a straight hydrostatic bursting test in accordance with Section 7.3.

7.4.2 Conditioned Hose Flexing Test

- 7.4.2.1 The 12-foot hose length marked "B" shall be installed in flexing test machine (Fig. 1). One end of the hose is to be connected to the traveling block and the free end passed around two pulleys with diameters as shown in Table 1. A 30-pound weight shall then be attached to the free end.
- 7.4.2.2 From the remainder of hose length marked "A", (sizes 1 inch and under only), cut a section to length indicated in Table 1. Connect one end to the vertically traveling block as shown in Fig. 1 and connect the other end to the liquid space of a tank of anhydrous ammonia. Maintain the temperature of hose and ammonia between 70°F and 90°F. The test on the feeder hose does not apply to sizes over 1 inch. To conduct the flex test on the larger sizes any convenient hose may be used as a feeder hose.
- 7.4.2.3 The flexing test shall continue for 72 hours at a rate of approximately 470 cycles per hour with a 42-inch vertical movement of the traveling block. A valve between the ammonia tank and the hose may be closed providing it is opened at least once each day to pressurize the hose. The hose shall be examined each day for visible defect. There shall be no evidence of cover blistering or leakage.
- 7.4.2.4 At the conclusion of the flexing period, cut a 24-inch sample from hose "A" and from hose "B" and subject each sample to a straight burst test in accordance with Section 7.3. All samples shall have a minimum burst of 1750 psig.

TABLE 1		
Hose Size	Pulley Diameter	Feeder Hose Length
½	14" ± ¼"	36"
¾"	14" ± ¼"	36"
1"	14" ± ¼"	36"
1 ¼"	15" ± ¼"	
1 ½"	18" ± ¼"	
2"	24" ± ¼"	

7.5 Low Temperature Test

A straight piece of hose at least 24 inches long, conditioned to minus 40°F plus or minus 2°F for 5 hours, and bent 180° within two seconds around a mandrel 12 times the nominal inside diameter of the hose, shall not break or show cracks in the tube or cover.

8. RETESTS AND REJECTIONS

Any hose which fails in one or more tests may be resampled and retested, for which purpose two additional samples shall be selected from the hose for the test that failed to meet the requirements. Failure of either of the retested samples shall be cause for final rejection.

9. HOSE ASSEMBLIES

The couplings must be so designed and constructed that an assembly shall have sufficient strength that it will reach the minimum burst pressure, as required by Section 4.3, before the end fittings leak or come off when pressure is applied as specified in ASTM D-380 for Hydrostatic Tests. Fittings must be resistant to the action of anhydrous and aqueous ammonia and in no case may assemblies be supplied with copper alloy fittings.

10. MARKINGS

Hose shall be clearly marked at least once every five feet with manufacturer's name or trademark, "Anhydrous Ammonia," the maximum working pressure in psig, year of manufacture, and "TFI-RMA Spec.," for all hose manufactured after January 1, 1964. As indicated in the Scope, the maximum working pressure must not be less than 350 psig.

11. PACKAGING

11.1 Packing

Unless otherwise specified, hose shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery to common or other carriers, at the lowest rate, to the point of delivery specified on the order.

11.2 Identification

Unless otherwise specified, shipping containers shall be marked with the size and quantity of hose therein, the name of the manufacturer and the number of the order.

Editor's Notes

History

Entire rule eff. 11/30/2008.

Entire rule eff. 12/30/2015.

Rules 1.3, 1.4.8, 1.4.18, 1.4.20, 2.6.1, 2.7.1, 3.1.2, 3.1.7, 6.4.1, 7.3.2, Section 8, 8.3.1, Section 9, 9.3.1, 11.3.4, 14.3 eff. 12/15/2020.